METRO TICKET GENERATING SYSTEM

Ideation Phase -- Brainstorm & Idea Prioritization Template

Date	**2 October 2026**
Team ID	
Project Name	Metro Ticket Generating System in ServiceNow Using Service Catalog and Workflows
Maximum Marks	**4 Marks**

Reference: https://www.mural.co/templates/brainstorm-and-idea-prioritization

Template Overview: Brainstorming provides a free and open environment that encourages everyone within a team to participate in the creative thinking process that leads to problem solving. Prioritizing volume over value, out-of-the-box ideas are welcome and built upon, and all participants are encouraged to collaborate, helping each other develop a rich amount of creative solutions.

STEP-1: TEAM GATHERING, COLLABORATION AND SELECT THE PROBLEM STATEMENT

- **Team Gathering:** Define who should participate in the session and share relevant information or pre-work ahead.
- **Set the Goal:** Focus on solving bottlenecks in commuter ticketing, manual pricing, and physical queue congestion.
- **Selected Problem Statement:** Metro ticket purchases and smart card management need an automated, digital process to collect travel details, calculate dynamic zone-based fares, generate digital QR-code passes, and route valid requests for station operator verification.

STEP-2: BRAINSTORM, IDEA LISTING AND GROUPING

- **Request Submission:** Create a ServiceNow Service Catalog Record Producer item (x_metro_ticket) to capture origin station, destination station, ticket type, travel class, and smart card ID.
- **Backend Pricing & Logic:** Implement server-side Script Includes (MetroFareCalculator) to compute distance- and zone-tier pricing automatically upon record submission.
- **Approval & Generation:** Generate a unique cryptographic QR code payload (u_qr_payload) upon booking validation and approval.
- **Fulfillment & Validation:** Create catalog tasks and equip station operators with a Configurable Workspace to scan and validate active passes.
- **Communication:** Send email notifications confirming successful bookings, rejections, or top-up receipts.
- **Exception Handling:** Use data validation rules to flag invalid route paths or insufficient smart card balances.

STEP-3: IDEA PRIORITIZATION (IMPORTANCE VS. FEASIBILITY)

Idea / Concept	Importance	Feasibility
Service Catalog request form (Record Producer)	High	High
Automated Script Include for zone-based fare calculation	High	High
Digital QR-code ticket payload generation	High	High
Station Operator Workspace validation view	High	High
Email notification confirmations	Medium	High
Multi-level managerial approval routing	Medium	Medium